

SIMATS ENGINEERING

ASSESSMENT TOOL 2

Case study

COURSE NAME: Computer Networks

COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO3: *Analyze the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions*

Assessment Tool Weightage: 40%

Case study

QUESTIONS:4

Total Marks:40

1.A multinational IT company uses a cloud-based video conferencing platform for daily client meetings. During peak business hours, employees experience frequent voice distortion, frozen video frames, and delayed communication. Network monitoring reports indicate an **8% packet loss, 180 ms jitter**, and fluctuating bandwidth utilization. The IT administrator suspects that the issue is related to the transport protocol or application-level buffering strategy.

Questions

1. Analyze whether the problem is primarily caused by the **transport protocol (TCP/UDP)** or the **application layer**.
2. Justify your answer based on the communication requirements of real-time multimedia applications.
3. Recommend suitable protocol-level or application-level improvements to enhance conferencing quality.

2.A multinational software company hosts its web services across multiple continents. Employees frequently report long delays before the application homepage loads. Network analysis shows that the **average DNS lookup time is approximately 800 ms**, significantly affecting user experience. The organization plans to redesign its DNS infrastructure while ensuring continuous service availability during server failures.

Questions

1. Analyze the reasons behind the high DNS resolution time.
2. Propose a suitable DNS architecture using techniques such as **Anycast DNS, DNS pre-fetching**, and **TTL optimization** to reduce the lookup time below **50 ms**.

3. Evaluate the advantages and limitations of your proposed architecture in terms of performance, scalability, and availability.
- 3.A cloud service provider delivers applications to millions of users worldwide. During promotional events, users experience intermittent delays in accessing cloud-hosted services. Investigation reveals that DNS queries are taking nearly **800 ms** because all requests are directed to a single primary DNS server. The company wants a highly available DNS solution capable of handling global traffic efficiently.

Questions

4. Examine the impact of centralized DNS architecture on application performance.
5. Design a distributed DNS solution that minimizes lookup latency while maintaining high availability.
6. Analyze how **TTL values**, **Anycast routing**, and **DNS caching** influence system performance and fault tolerance.
- 4.A financial institution operates an electronic stock trading platform where thousands of buy and sell orders are submitted every second. During market opening hours, the **average order acknowledgment latency increases from 1 ms to 40 ms**, resulting in delayed trade confirmations and customer dissatisfaction. Network engineers observe increased TCP retransmissions, longer connection establishment times, and excessive buffering.

Questions

1. Analyze three possible TCP-related factors responsible for the latency increase, considering **flow control**, **Nagle's algorithm**, and **SYN queue limitations**.
2. Explain how each factor affects transaction processing in high-frequency systems
3. Recommend appropriate TCP configuration changes to reduce latency and improve transaction reliability.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided

Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand
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Case Study 1

Question 1

Analyze whether the problem is primarily caused by the transport protocol (TCP/UDP) or the application layer.

The problem is mainly related to the **transport protocol (UDP)**, but the **application layer** also contributes to the issue. Video conferencing applications usually use UDP because it provides faster communication without waiting for retransmissions. However, when packet loss reaches **8%** and jitter is around **180 ms**, voice and video quality are affected, resulting in frozen video, distorted audio, and communication delays. The application layer may also cause delays if its buffering mechanism is not properly configured to handle network fluctuations.

Question 2

Justify your answer based on the communication requirements of real-time multimedia applications.

Real-time applications such as video conferencing require **low latency, low jitter, and continuous data transmission**. UDP is preferred because it sends data without waiting for acknowledgements, allowing audio and video to arrive quickly. Although some packets may be lost, real-time communication is maintained. Using TCP would increase delays because lost packets must be retransmitted, making conversations appear slow and unnatural. Therefore, UDP is more suitable, provided that the network has low packet loss and stable bandwidth.

Question 3

Recommend suitable protocol-level or application-level improvements to enhance conferencing quality.

To improve conferencing quality, the organization should reduce packet loss and jitter by implementing **Quality of Service (QoS)** to prioritize voice and video traffic. The application can use **adaptive jitter buffers** that automatically adjust according to network conditions. Modern codecs that support error recovery can also improve audio and video quality. Increasing available bandwidth, reducing network congestion, and using **Forward Error Correction (FEC)** can further improve communication without introducing large delays. These improvements will provide smoother video, clearer voice, and a better user experience.

Case Study 2

Question 1

Analyze the reasons behind the high DNS resolution time.

The main reason for the high DNS lookup time of **800 ms** is that the DNS servers are not located close to the users. Every DNS request has to travel a long distance before receiving a response, which increases the delay. The DNS server may also

be overloaded because it is handling requests from multiple continents. Poor DNS caching, incorrect TTL values, and network congestion can further increase the lookup time. As a result, users experience a noticeable delay before the website begins to load.

Question 2

Propose a suitable DNS architecture using techniques such as Anycast DNS, DNS pre-fetching, and TTL optimization to reduce the lookup time below 50 ms.

A better solution is to use a **distributed Anycast DNS architecture**. In this setup, multiple DNS servers are placed in different geographic locations but share the same IP address. User requests are automatically directed to the nearest DNS server, reducing response time.

DNS pre-fetching can also be enabled so that frequently visited domain names are resolved before the user requests them. This reduces the waiting time while loading web pages.

TTL values should be set to an optimal level. A moderate TTL allows DNS records to remain in cache for an appropriate amount of time while still updating changes when necessary. Combining Anycast DNS, DNS caching, and optimized TTL values can reduce DNS lookup time to well below **50 ms**.

Question 3

Evaluate the advantages and limitations of your proposed architecture in terms of performance, scalability, and availability.

The proposed architecture offers several advantages. Anycast DNS improves performance by directing users to the nearest DNS server, which significantly reduces latency. Since multiple DNS servers are available, the system can handle a large number of requests, making it highly scalable. It also provides high availability because if one DNS server fails, traffic is automatically redirected to another server without affecting users.

However, the architecture also has a few limitations. Setting up and maintaining multiple DNS servers requires higher cost and careful network management. DNS records must be synchronized across all servers to ensure consistent responses.

Despite these challenges, the benefits of improved speed, scalability, and reliability make this architecture suitable for large multinational organizations.

Case Study 3

Question 1

Examine the impact of centralized DNS architecture on application performance.

In a centralized DNS architecture, all DNS requests are handled by a single primary DNS server. When the number of users increases, especially during promotional events, the server becomes overloaded and takes longer to respond. This increases the DNS lookup time and delays the loading of cloud-hosted applications. If the primary DNS server fails, users may not be able to access the application at all. Therefore, a centralized DNS system reduces performance, increases latency, and creates a single point of failure.

Question 2

Design a distributed DNS solution that minimizes lookup latency while maintaining high availability.

A distributed DNS architecture should be used instead of a single DNS server. Multiple DNS servers can be deployed in different geographic regions and connected using **Anycast routing**. Since all the DNS servers share the same IP address, user requests are automatically directed to the nearest available server. DNS caching should also be enabled so that frequently accessed domain names are stored locally. This reduces the number of repeated DNS lookups. In addition, backup DNS servers should be configured to provide continuous service if one server becomes unavailable.

With this setup, users receive faster responses, server load is shared among multiple DNS servers, and the network remains available even if one server fails.

Question 3

Analyze how TTL values, Anycast routing, and DNS caching influence system performance and fault tolerance.

TTL (Time to Live) determines how long a DNS record is stored in the cache. A suitable TTL value reduces the number of DNS queries while still allowing updates to be reflected within a reasonable time. If the TTL is too low, DNS servers receive more requests. If it is too high, changes to DNS records may take longer to reach users.

Anycast routing improves performance by directing DNS requests to the nearest available DNS server. This reduces lookup time, balances traffic, and automatically redirects users if one server becomes unavailable.

DNS caching stores recently accessed DNS records in local devices or DNS resolvers. This avoids repeated lookups for the same domain name, reducing network traffic and improving response time.

By combining appropriate TTL values, Anycast routing, and DNS caching, the DNS infrastructure becomes faster, more reliable, and capable of handling a large number of users while maintaining high availability.

Case Study 4

Question 1

Analyze three possible TCP-related factors responsible for the latency increase, considering flow control, Nagle's algorithm, and SYN queue limitations.

The increase in latency from **1 ms to 40 ms** can be caused by several TCP-related factors. The first is **flow control**. If the receiver cannot process incoming data quickly, TCP reduces the transmission rate, causing delays in sending acknowledgements.

The second factor is **Nagle's algorithm**. This algorithm combines small packets into a larger packet before transmission to reduce network congestion. Although this improves bandwidth utilization, it introduces additional delay, which is not suitable for high-frequency trading applications.

The third factor is **SYN queue limitations**. During market opening hours, thousands of new TCP connections are created at the same time. If the SYN queue

becomes full, new connection requests are delayed or dropped, increasing the connection establishment time.

Question 2

Explain how each factor affects transaction processing in high-frequency systems.

In high-frequency trading systems, every millisecond is important. If flow control slows down data transmission, trade orders reach the server later than expected.

This may result in delayed order execution.

Nagle's algorithm delays the transmission of small packets while waiting to combine them with other packets. Since stock trading applications send many small messages, this delay affects the speed of trade confirmations.

A full SYN queue delays the creation of new TCP connections. During periods of heavy traffic, traders may experience slow logins, delayed order submissions, or even failed connections. These problems reduce the overall performance and reliability of the trading platform.

Question 3

Recommend appropriate TCP configuration changes to reduce latency and improve transaction reliability.

To reduce latency, **Nagle's algorithm should be disabled** using the

TCP_NODELAY option so that small packets are transmitted immediately. The **TCP receive and send buffer sizes** should be adjusted to improve flow control and avoid unnecessary delays. The **SYN queue size** should also be increased so that the server can handle a large number of connection requests during peak trading hours.

In addition, high-speed network hardware, Quality of Service (QoS), and continuous network monitoring should be used to reduce congestion and improve reliability. These changes help ensure faster order processing, lower latency, and a better trading experience for customers.